Algebra Review: Unit 1 — Foundations of Algebra

Station 1 of 4: Order of Operations and Evaluating Expressions

Name: _____ Date: _____

Show your work. Simplify completely.

1. Evaluate: $12 - 3 \cdot 2 + 5$

2. Evaluate: $(8 + 4) \div 3 \cdot 2$

3. Evaluate: $5 + 2^3 - 4$

4. Evaluate: $36 \div (2 + 4) - 1$

5. Evaluate: $2[10 - (3 + 4)]$

6. Evaluate: $\dfrac{15 - 3^2}{2 \cdot 3}$

7. Evaluate $3x + 4$ for $x = -2$.

8. Evaluate $x^2 - 5x$ for $x = 3$.

9. Evaluate $\dfrac{a + b}{2}$ for $a = 7$ and $b = -1$.

10. Evaluate $2m^2 - n$ for $m = -3$ and $n = 5$.

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Station 2 of 4: Real Numbers and Properties

Name: _____ Date: _____

List every set each number belongs to: natural, whole, integer, rational, irrational. Then name each property.

1. Classify -6.

2. Classify 0.

3. Classify $\sqrt{49}$.

4. Classify $\frac{2}{3}$.

5. Classify $\sqrt{10}$.

6. Classify -0.25.

7. Name the property: $7 + 9 = 9 + 7$

8. Name the property: $(3 \cdot 4) \cdot 5 = 3 \cdot (4 \cdot 5)$

9. Name the property: $8 + 0 = 8$

10. Name the property: $5(x - 2) = 5x - 10$

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Station 3 of 4: Combining Like Terms and the Distributive Property

Name: _____ Date: _____

Simplify each expression.

1. $7x + 3x$

2. $4a - 9 + 2a + 5$

3. $3y^2 + 2y - y^2 + 6y$

4. $5(x + 6)$

5. $-3(2n - 4)$

6. $2(3x + 1) + 4x$

7. $6 - (x + 4)$

8. $4(2a - 1) - 3(a + 5)$

9. $\frac{1}{2}(8x - 6) + 3$

10. $3(2x - y) - 2(x + 4y)$

Algebra Review: Unit 1 — Foundations of Algebra

Station 4 of 4: Words to Expressions and Applications

Name: _____ Date: _____

Write an expression. When a value is given, evaluate.

1. The sum of a number n and 9

2. 7 less than a number x

3. The product of 6 and a number y

4. Twice a number x, decreased by 5

5. The quotient of a number m and 4, increased by 1

6. Three times the sum of a number x and 2

7. A movie ticket costs \$12 and popcorn costs \$5. Write an expression for the cost of t tickets and one popcorn.

8. Use your expression from #7 to find the cost of 4 tickets and one popcorn.

9. A rectangle is $x + 3$ long and 4 wide. Write and simplify an expression for its perimeter.

10. A phone plan costs \$20 plus \$0.10 per text. Write an expression for t texts, then find the cost of 150 texts.